

BUDDY FOR YOUTH (B-YOUTH)

Career Recommendation Berbasis Artificial Intelligence dengan Integrasi Konseling Psikologi

Mid-Term Assessment (ATS) Project-Based Learning
Politeknik Negeri Batam · Teknik Informatika · IF 3D Pagi
Team Code: PBLIF3-PD09

MEET OUR TEAM

TEAM LEADER



Muhammad Faturrahman
3312501043
DEVELOPER

Technical coordination, backend development, system integration, and repository management.

Project Manager

Sukma Evadini, S.T., M.Kom.

Directing and supervising project implementation; monitoring progress and evaluating team results.

TEAM MEMBER



Nur Iliyanie
3312501045
DEVELOPER

Requirements analysis, UI design, testing, and documentation.

Why B-Youth is Needed?

01

Limited information regarding study programs and career prospects.

02

Access to psychological counseling is still limited.

03

Difficulty in matching academic grades with suitable study options.

04

Lack of understanding regarding personal interests, talents, and potential.

B-Youth Development Objectives

Student — Assessment — AI Matching — Study Program Recommendation — Psychological Counseling

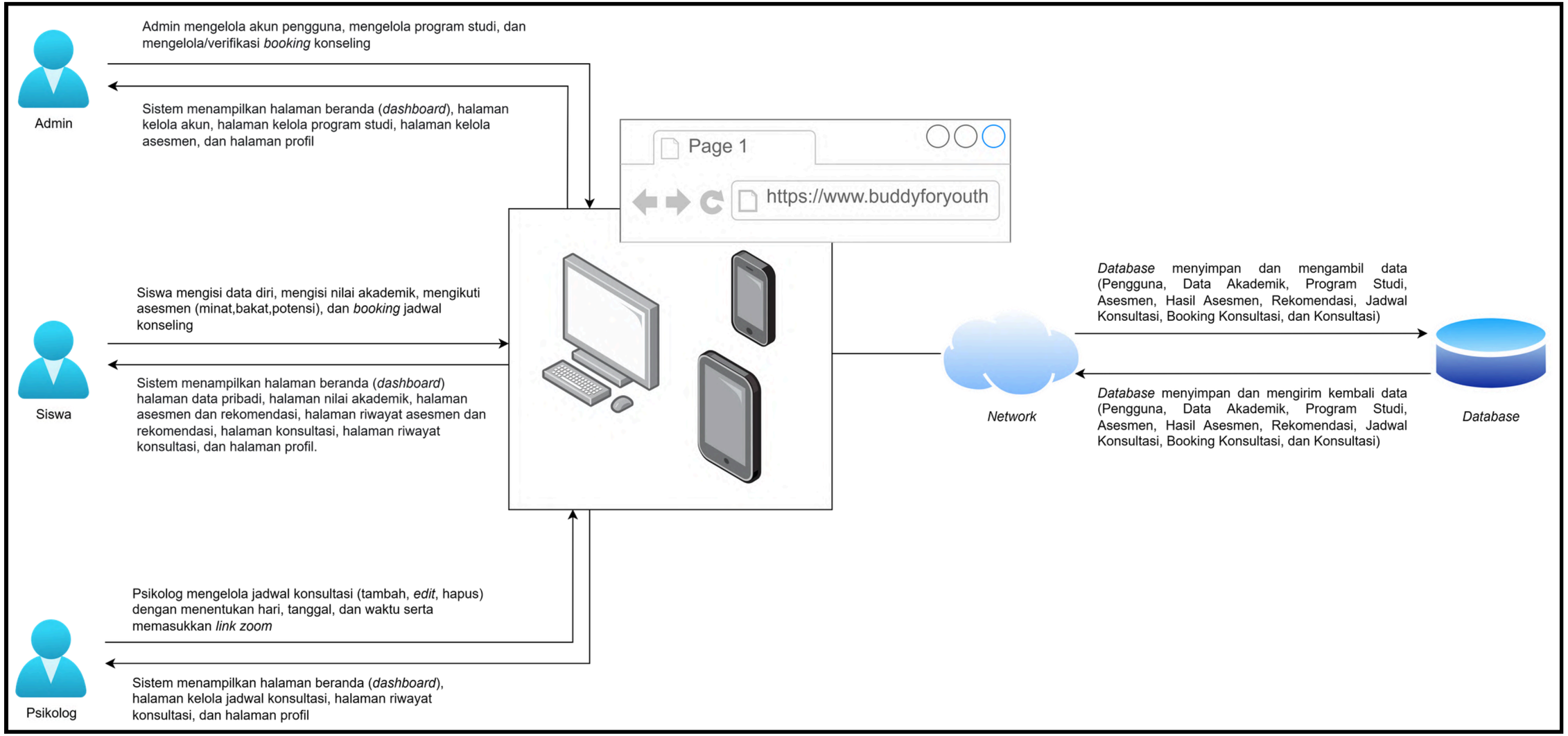
01
Identify interests, talents, potential, and personal preferences.

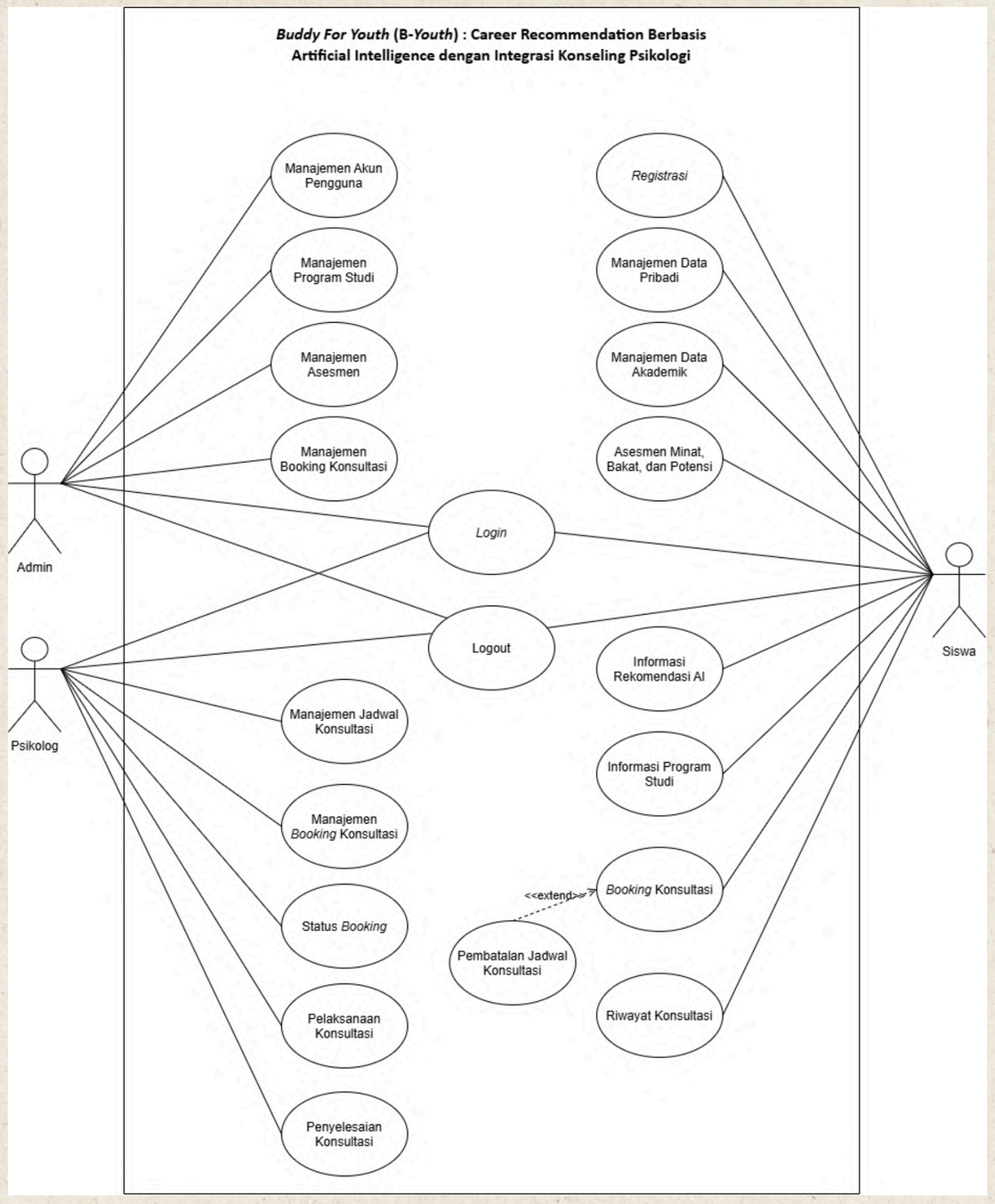
02
Combine assessment results with academic performance/grades.

03
Understand study programs and future career prospects.

04
Access psychological counseling services.

System Overview





Use Case Diagram

How does B-Youth work?

01
Registration / Login

Students sign in to the platform.

02
Data Entry

Entering personal information and academic grades.

03
Assessment

Evaluating interests, talents, potential, and preferences.

04
AI Processing

Recommendation algorithms process the data.

05
Recommendation Results

Recommended study programs, match scores, rationales, and career prospects.

06
Psychological Counseling

Request and schedule a consultation session.

AI recommendations are a consideration guide, not an absolute decision.

Technology and Artificial Intelligence Methods

DEVELOPMENT TECHNOLOGIES

- **Laravel**
- **PHP**
- **MySQL**
- **Tailwind CSS**
- **JavaScript**

01

Cosine Similarity
Calculates similarity between student profiles and study programs.

02

Forward Chaining
Applies IF-THEN rule-based reasoning derived from assessments and grades.

03

Simple Additive Weighting (SAW)
Combines multi-criteria attributes to generate scores and rankings.

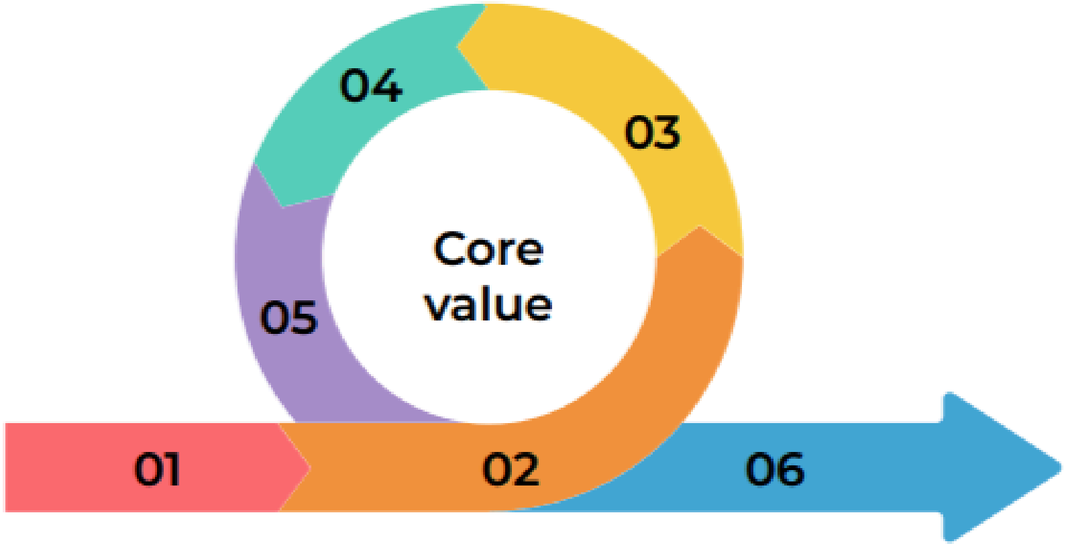
04

Text Generation
Produces contextual explanations and descriptive recommendations.

STUDENT DATA → WEIGHTING & RULES → MATCH SCORE → RECOMMENDATION

Development Methodology

	Plan		Test
	Design		Review
	Develop		Deploy



Perkembangan Pengerjaan Proyek

SPRINT 0

- Initial meeting with project manager and client.
- Discussion of overall project scope.
- Identification of system requirements.

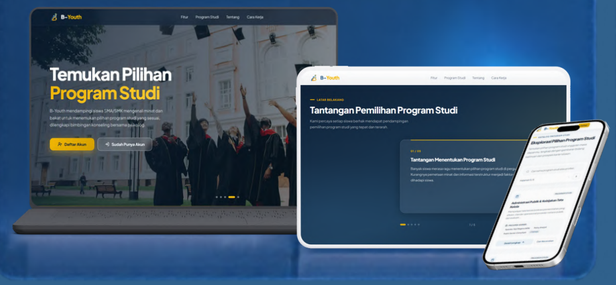
SPRINT 1

- US-01: Login
- US-02: Registration
- US-03: Account Management
- US-04: Study Program Management
- US-05: Assessment Management
- US-06: Admin Dashboard
- US-07: Student Dashboard
- US-08: Student Academic Data
- EN-01: Preparation of AI Match Dataset

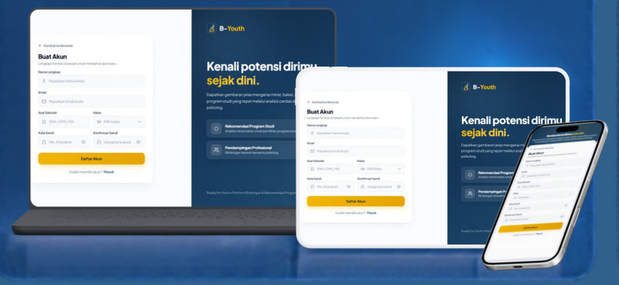
UI/UX PLATFORM

B-YOUTH

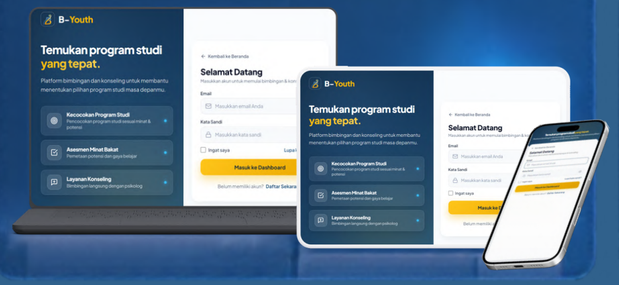
1 Landing Page



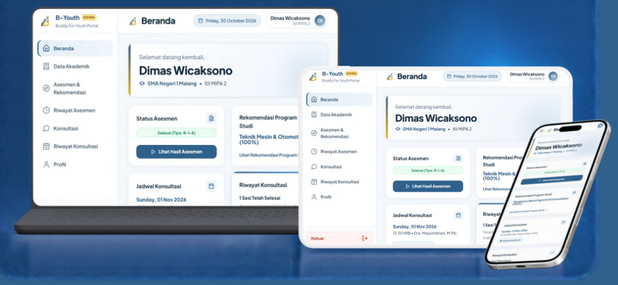
2 Registration



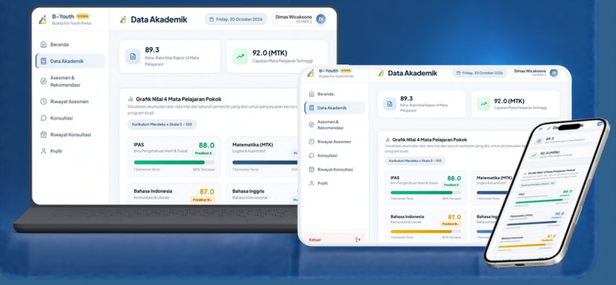
3 Login



4 Student Dashboard



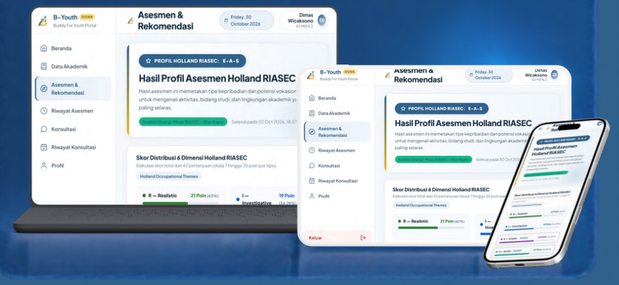
5 Academic Page



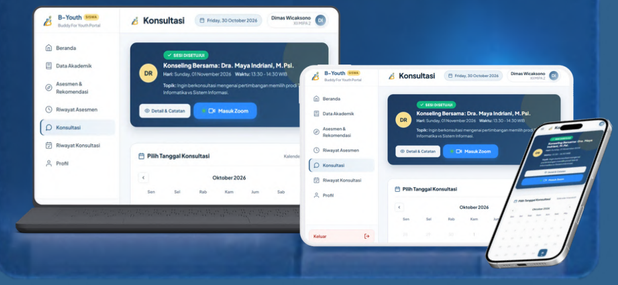
6 Assessment Page



7 Recommendation Page



8 Consultation Page



THANK YOU

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Politeknik Negeri Batam

